

DIFFERENTIAL EQUATIONS

Lecture 8 — Orthogonal Trajectories

Definition

Definition: Orthogonal Trajectories

Two families of curves are called **orthogonal trajectories** of each other if every curve from one family intersects every curve from the other family at **right angles** (90°) at every point of intersection.

The Perpendicularity Condition

Condition: Two curves are perpendicular when...

If two curves meet at a point with slopes m_1 and m_2 , they are **perpendicular** if and only if:

$$m_1 \cdot m_2 = -1 \quad \implies \quad m_2 = -\frac{1}{m_1}$$

So if the original family has slope $dy/dx = F(x, y)$ at every point, the orthogonal trajectories must satisfy the new DE obtained by replacing F with its negative reciprocal.

Method

Steps to Find the Orthogonal Trajectories of $f(x, y, c) = 0$

1. **Find the DE of the original family.** Differentiate $f(x, y, c) = 0$ w.r.t. x and eliminate c to get $\frac{dy}{dx} = F(x, y)$.
2. **Apply the perpendicularity condition.** Replace $\frac{dy}{dx}$ with $-\frac{1}{F(x, y)}$.
3. **Solve the new DE** to find the family of orthogonal trajectories $g(x, y, k) = 0$.

If c does not disappear after differentiating, eliminate it algebraically using the original equation before writing $dy/dx = F(x, y)$.

Worked Examples

Example 1: OTs of $x^2 - y^2 = c$ (hyperbolas)

Step 1. Differentiate $x^2 - y^2 = c$:

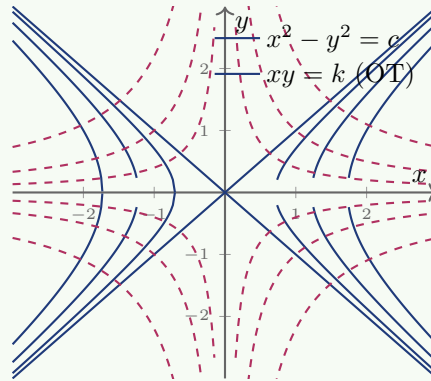
$$2x - 2y \frac{dy}{dx} = 0 \quad \implies \quad \frac{dy}{dx} = \frac{x}{y}$$

Step 2. Replace slope:

$$\frac{dy}{dx} = -\frac{y}{x}$$

Step 3. Separate and integrate:

$$\frac{dy}{y} = -\frac{dx}{x} \implies \ln|y| = -\ln|x| + C \implies \boxed{xy = k}$$



Example 2: OTs of $y = x^2 + c$ (vertical parabolas)

Step 1. Differentiate $y = x^2 + c$:

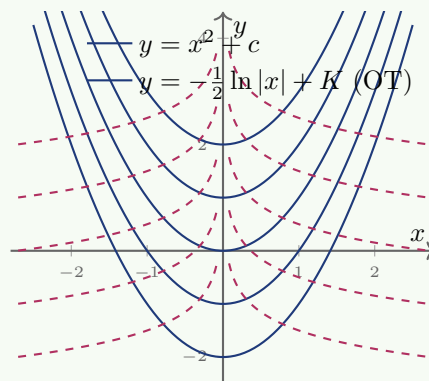
$$\frac{dy}{dx} = 2x$$

Step 2. Replace slope:

$$\frac{dy}{dx} = -\frac{1}{2x}$$

Step 3. Integrate directly:

$$\int dy = \int -\frac{1}{2x} dx \implies \boxed{y = -\frac{1}{2} \ln|x| + K}$$



Example 3: OTs of $y^2 = 4cx$ (horizontal parabolas)

Step 1. Eliminate c : from $y^2 = 4cx \Rightarrow c = y^2/(4x)$. Differentiate:

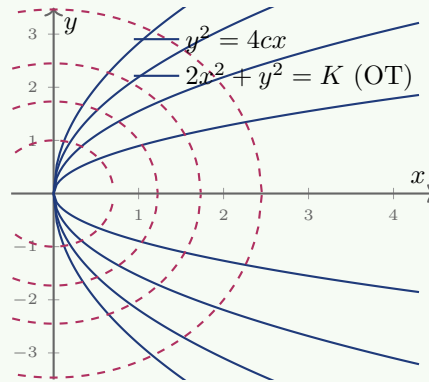
$$2y \frac{dy}{dx} = 4c = \frac{y^2}{x} \implies \frac{dy}{dx} = \frac{y}{2x}$$

Step 2. Replace slope:

$$\frac{dy}{dx} = -\frac{2x}{y}$$

Step 3. Separate and integrate:

$$y \, dy = -2x \, dx \implies \frac{y^2}{2} = -x^2 + K \implies \boxed{2x^2 + y^2 = K}$$



Example 4: OTs of $x^2 + y^2 = c^2$ (concentric circles)

Step 1. Differentiate:

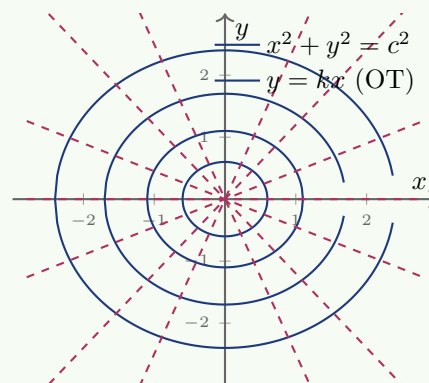
$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}$$

Step 2. Replace slope:

$$\frac{dy}{dx} = \frac{y}{x}$$

Step 3. Separate and integrate:

$$\frac{dy}{y} = \frac{dx}{x} \implies \ln |y| = \ln |x| + C \implies \boxed{y = kx}$$



Example 5: OTs of $y = ce^x$ (exponential curves)

Step 1. Differentiate: $dy/dx = ce^x$. Eliminate c using $c = ye^{-x}$:

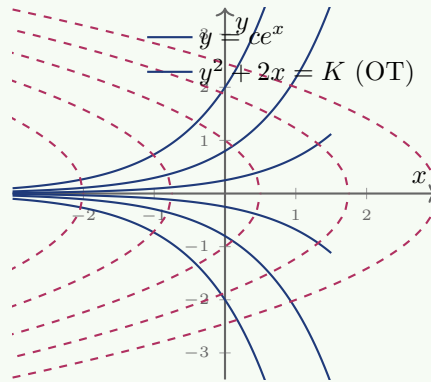
$$\frac{dy}{dx} = ye^{-x} \cdot e^x = y \implies \frac{dy}{dx} = y$$

Step 2. Replace slope:

$$\frac{dy}{dx} = -\frac{1}{y}$$

Step 3. Separate and integrate:

$$y \, dy = -dx \implies \frac{y^2}{2} = -x + K \implies \boxed{y^2 + 2x = K}$$



Summary

Concept	Key Point
Orthogonal trajectories	Two families cutting each other at 90° everywhere
Perpendicularity condition	$m_1 \cdot m_2 = -1 \Rightarrow m_2 = -1/m_1$
Step 1	Differentiate and eliminate c to get $dy/dx = F(x, y)$
Step 2	Replace dy/dx with $-1/F(x, y)$
Step 3	Solve the new DE (separable, exact, or linear as needed)
If c remains	Eliminate it from the original equation first

End of Lecture 8